Tutorial 2

Tensors

26 September 2025

Throughout this tutorial, let $(V, +, \cdot)$ be an n-dimensional real vector space.

1. Give the definition of a $\binom{p}{q}$-tensor on V.

   Give the definition of the components of a $\binom{p}{q}$-tensor.

   Solution: See slides and lecture notes.

2. Tick the true statements:

   - [ ] The number of indices of the tensor components depends on the dimension of the vector space.
   - [x] A change of basis in a vector space does not change a tensor in that space.
   - [x] Knowing a basis of V is sufficient to determine tensor components.
   - [x] The map that takes a vector in V and returns its first component is a $\binom{0}{1}$-tensor.
   - [x] For any $v \in V$ and $\alpha \in V^*$, the number $\alpha_i\, v^i$ does not depend on the choice of basis.
   - [ ] Having chosen a basis in a vector space one can calculate angles between vectors.
   - [x] The map $\varphi\colon V \to \mathbb{R};\ v \mapsto \varphi(v) := 0$ is the zero element of the dual space V^*.
   - [ ] A bilinear map $\kappa\colon V \times V \to \mathbb{R}$ satisfies $\kappa(\lambda \cdot v, \lambda \cdot w) = \lambda\, \kappa(v, w)$.
   - [x] Any linear map $\varphi\colon V \to V$ can be regarded as a $\binom{1}{1}$-tensor.

3. Show the one-to-one correspondence between linear maps $\varphi\colon V \to V$ and $\binom{1}{1}$-tensors on V. Consider the following steps:

   - Given a $\binom{1}{1}$-tensor $T\colon V^* \times V \to \mathbb{R}$, construct a corresponding linear map $\varphi_T\colon V \to V$.
   - Give the definition of components of the linear map $\varphi_T\colon V \to V$ with respect to a basis $\{e_1, \ldots, e_N\}$ of V.
   - Check that the latter components coincide with the components of the $\binom{1}{1}$-tensor from which φ_T is defined.

Since components completely describe the abstract object they define, these steps prove the equivalence of a linear map with $\binom{1}{1}$-tensors.

> **Solution:**
> $$\forall \omega \in V^* : \quad \omega(\varphi_T(v)) = T(\omega, v)$$
> $$\varphi_T(e_j) := \varphi^i_j e_i$$
> $$\epsilon^i(\varphi_T(e_j)) = \epsilon^i(\varphi^k_j e_k) = \varphi^i_j = T^i_j$$

4. Consider a vector space V and two different basis for it $\{e_1, \dots, e_N\}$ and $\{\tilde{e}_1, \dots, \tilde{e}_N\}$ and $\{e_1, \dots, e_N\}$, with corresponding dual basis $\{\epsilon^1, \dots, \epsilon^N\}$ and $\{\tilde{\epsilon}^1, \dots, \tilde{\epsilon}^N\}$, so that a vector $v \in V$ can be described as $v = v^i e_i = \tilde{v}^i \tilde{e}_i$ and a covector $\alpha \in V^*$ as $\alpha = \alpha_i \epsilon^i = \tilde{\alpha}_i \tilde{\epsilon}^i$. A matrix A defines the tranformation rule between basis, i.e, $\tilde{e}_j = A^i_j e_i$.

   - Understand deeply the index notation convention (why and which indices are up/down?) in relation to einstein summation convention.
   - Derive the transformation rule for the components of vectors and co-vectors and for the dual basis, i.e., calculate how $\tilde{v}^i$ and v_i, $\tilde{\alpha}_i$ and α_i, and finally $\tilde{\epsilon}^i$ and ϵ^i are related. The result should clarify why co-vector components are called *covariant* and vector components *contravariant*.
   - Derive the transformation rule of general tensor components given the definition of components of tensors.

> **Solution:** see lecture notes

5. Consider a linear map $\varphi \colon V \to W$. Give the definition of the dual map φ^*.

   Let $(e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$ be bases of V and W, respectively. Show that the components of φ^* coincide with the components of φ, i.e. $(\varphi^*)^j{}_i = \varphi^j{}_i$.

> **Solution:** See the lecture notes.

⚠ The expression $\varphi^*(\beta) = \alpha$ is written in components as $(\varphi^*)^j{}_i \, \beta_j = \alpha_i$. Use of the witchcraft matrix multiplication might bring confusion! Stacking components of covectors into rows, gives

$$\boldsymbol{\beta} \cdot \Phi^* = \boldsymbol{\alpha} \qquad \text{or} \qquad (\Phi^*)^\top \cdot \boldsymbol{\beta}^\top = \boldsymbol{\alpha}^\top.$$

Using the proven equality $(\varphi^*)^j{}_i = \varphi^j{}_i$ (or $\Phi^* = \Phi$), we get

$$\boldsymbol{\beta} \cdot \Phi = \boldsymbol{\alpha} \qquad \text{or} \qquad \Phi^\top \cdot \boldsymbol{\beta}^\top = \boldsymbol{\alpha}^\top.$$

Transposition is merely a rewriting of expressions in components, not a real operation!

6. Given a vector space V, is the dual space V^* unique?

> **Solution:** Yes.

Given a vector $v \in V$, is there a unique covector $v^* \in V^*$ such that $v^*(v) = 1$?

> **Solution:** No.

Now, equip V with an inner product $g\colon V \times V \to \mathbb{R}$.

7. Consider $V = \mathbb{R}^2$ with the standard vector space structure.

   (a) Show that the map

   $$g\big((a,b),(c,d)\big) = ac + bc + ad + 4bd$$

   is bilinear and defines an inner product on $\mathbb{R}^2$.

   (b) What are the components g_{ij} of the inner product in the standard basis $\big((1,0),(0,1)\big)$ of $\mathbb{R}^2$? Is the standard basis orthonormal with respect to g?

   (c) Is the basis $\big((-1,\frac{1}{2}),(\frac{1}{\sqrt{3}},\frac{1}{2\sqrt{3}})\big)$ orthonormal with respect to g?

   (d) What are the components $\tilde{g}_{kl}$ of the inner product in this basis?

   (e) Provide the transformation rule for the components of an inner product (you can use the transformation rule of a $\binom{0}{2}$-tensor calculated in previous exercise) and verify that it holds for g_{ij} and $\tilde{g}_{kl}$ calculated above.

   (f) Write the transformation of the inner product components using matrix multiplication.

> **Solution:**
>
> (a) An inner product satisfies:
>
> (i) Symmetry:
>
> $$g\big((a,b),(c,d)\big) = ac + bc + ad + 4bd$$
> $$g\big((c,d),(a,b)\big) = ca + cb + da + 4db$$
> $$g\big((a,b),(c,d)\big) = g\big((c,d),(a,b)\big)$$
>
> (ii) Bilinearity:
>
> $$g\big(\lambda(a,b),(c,d)\big) = (\lambda ac + \lambda bc + \lambda ad + 4\lambda bd) = \lambda(ac + bc + ad + 4bd)$$
>
> $$g(\lambda \cdot u, v) = \lambda \cdot g(u,v)$$
>
> The same argument works for the second (c,d) slot.

(iii) Positive definiteness:

$$g((a,b),(a,b)) = a^2 + 2ab + 4b^2 > 0 \quad \forall (a,b) \neq (0,0)$$

$$(a,b) = (0,0) \implies g((a,b),(a,b)) = 0$$

(b)

$$e_1 = (1,0), e_2 = (0,1)$$
$$g_{11} = g(e_1,e_1) = g((1,0),(1,0)) = 1$$
$$g_{12} = g(e_1,e_2) = g((1,0),(0,1)) = 1$$
$$g_{21} = g(e_2,e_1) = g((0,1),(1,0)) = 1$$
$$g_{22} = g(e_2,e_2) = g(0,1),(0,1)) = 4$$
$$g_{22} \neq 1: \quad \text{not normalized}$$
$$g_{12} \neq 0: \quad \text{not orthogonal}$$
$$\text{Therefore not orthonormal}$$

(c) Filling in $\tilde{e}_1 = ((-1, \frac{1}{2})),\ \tilde{e}_2 = (\frac{1}{\sqrt{3}}, \frac{1}{2\sqrt{3}})$ gives:

$$g(\tilde{e}_1,\tilde{e}_1) = g(\tilde{e}_2,\tilde{e}_2) = 1, \quad g(\tilde{e}_1,\tilde{e}_2) = g(\tilde{e}_2,\tilde{e}_1) = 0$$

Therefore this basis is orthonormal with respect to g.

(d)

$$\tilde{g}_{11} = \tilde{g}_{22} = 1, \quad \tilde{g}_{21} = \tilde{g}_{12} = 0$$

(e) Basis changes as:

$$\tilde{e}_j = \sum_i A_i^j e_i$$

Components of the metric tensor change as:

$$\tilde{g}_{kl} = \sum_{i,j} A_k^i A_l^j g_{ij}$$

(f)

$$G := \begin{pmatrix} g_{11}, g_{12} \\ g_{21}, g_{22} \end{pmatrix}, \quad \tilde{G} := \begin{pmatrix} \tilde{g}_{11}, \tilde{g}_{12} \\ \tilde{g}_{21}, \tilde{g}_{22} \end{pmatrix}, \quad A := (\tilde{e}_1, \tilde{e}_2) = \begin{pmatrix} -1, \frac{1}{\sqrt{3}} \\ \frac{1}{2}, \frac{1}{2\sqrt{3}} \end{pmatrix},$$

$$\tilde{G} = A^T G A$$